

A07 OPERABLE WINDOWS | O (MAX : 2 PT)

Intent : Increase the supply of high-quality outdoor air and promote a connection to the outdoor environment, by encouraging building users to open windows when outdoor air quality is acceptable.

Summary : This WELL feature requires buildings with operable windows to increase the supply of high-quality outdoor air and promote a connection to the outdoor environment, by encouraging building users to open windows when outdoor air quality is acceptable.

Issue : Inhalation of harmful indoor substances is correlated with adverse health outcomes, such as infectious diseases, cancer, asthma, allergies and other respiratory infections.^{1,2} When outdoor air quality is favorable, opening windows can provide a supply of outdoor air and lower levels of indoor air contaminants. Achieving natural ventilation through open windows can provide a positive human experience.³ However, it is challenging to the ability to maintain strict control over interior air quality. Despite often having conditions of temperature and ventilation below recommended standards, studies have shown that naturally ventilated buildings have fewer people reporting sick building syndrome symptoms than mechanically conditioned buildings.^{4,5} In addition, studies have shown that there is a 7.7% productivity benefit associated with naturally ventilated spaces.^{5,6}

Solutions : When weather and local outdoor parameters indicate high-quality outdoor air, people should be encouraged to make use of natural ventilation strategies. However, ventilating through windows generally introduces more outdoor pollution than mechanical systems with filters.⁷ Human behavior is rated as one of the top three components affecting the generation and persistence of indoor air pollutants.⁷ Educating and informing building users when outdoor conditions are favorable for window opening, can reduce the generation and persistence of indoor air pollutants.

PART 1 PROVIDE OPERABLE WINDOWS (MAX : 1 PT)

Note :

Projects where the average outdoor PM_{2.5} level is 35 µg/m³ or higher may only achieve Part 1: Provide Operable Windows in conjunction with Part 2: Manage Window Use.

For All Spaces:

Project meets one of the below:

- At least 75% of the regularly occupied spaces have operable windows that provide access to outdoor air.
- For each floor, the openable window area is at least 4% the area of the indoor occupiable space.

Note : The World Health Organization's Global Urban Ambient Air Pollution Database may be consulted to view outdoor air quality levels, available at <https://www.who.int/data/gho/data/themes/air-pollution/who-air-quality-database>

PART 2 MANAGE WINDOW USE (MAX : 1 PT)

Note :

Projects may only achieve this part if Part 1 is also achieved.

For All Spaces:

1: Outdoor air measurement

The following requirement is met:

- Outdoor levels of PM_{2.5}, temperature and humidity are monitored at intervals of at least once per hour, based on a data-gathering station located within 4 km of the building. This monitoring system may be operated by the project or by another entity (e.g., a government).

2: Window operation

Indicator lights and/or digital displays at windows (at least one per room with windows) cue occupants when conditions outside are suitable for opening windows:

- PM_{2.5}: 15 µg/m³ or lower.
- Dry-bulb temperature: within 8 °C of indoor air temperature setpoint.
- Relative Humidity: 65% or lower.

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